

UTM TENSILE REPORT — UTM_Test_20260822_180547

DIC gauge strain (ENGINEERING: $\Delta L/L_0$) · engineering stress (F/A_0) · anchor self-calibration

41.7 MPa

UTS

31.5 MPa

Yield σ_y

1.81 GPa

Modulus E

7.8 %

Failure ϵ_f

2715 kJ/m³

Toughness

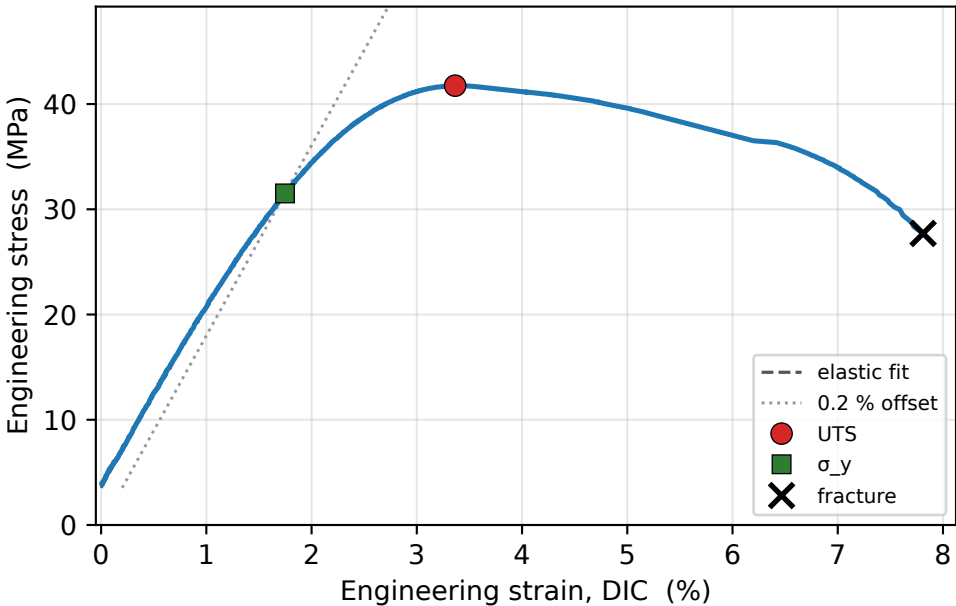
311 N

Preload anchor

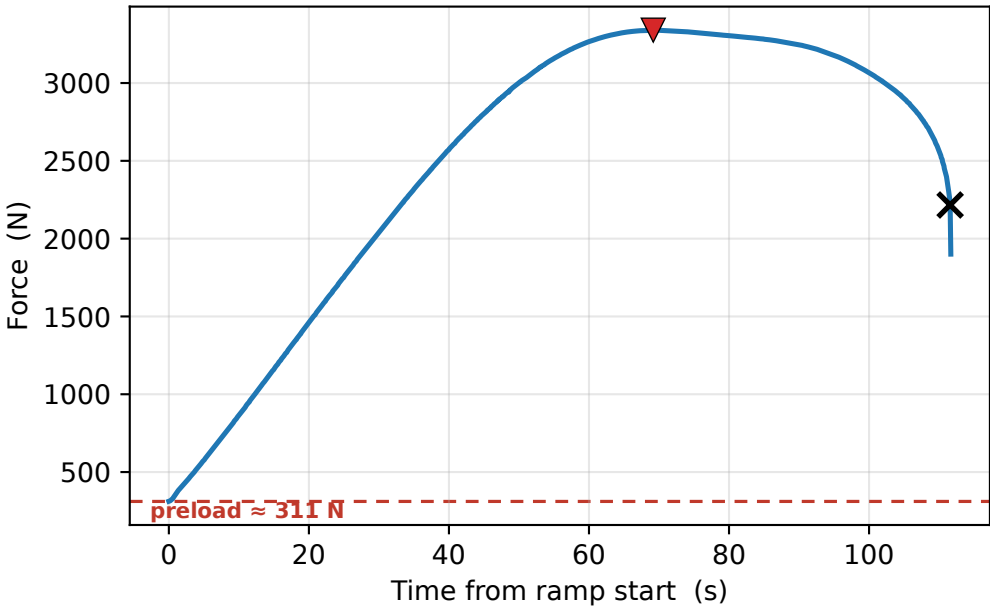
TEST SETTINGS

Specimen	UTM_Test_20260822_180547
DIC mode	Black
Preload target	300 N
Test speed	0.100 mm/s
Cross-section	80 mm ²
Gauge length L ₀	80 mm
Infill (label)	100 %
Force calib	sc -0.0065 / off 336.18
Comment	—
Test date	2026-08-22 18:02:30
Duration	201.1 s

Stress-strain



Load vs time

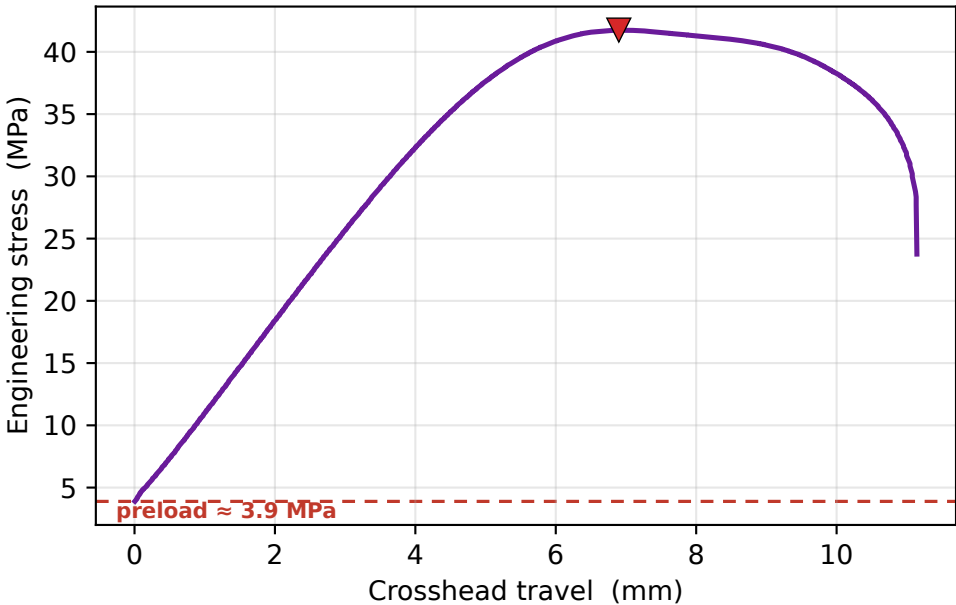


VALIDATION vs REFERENCES

prop	meas	Chacón	E-PLA k
UTS	41.7	32-60 PASS	×1.39
σ_y	31.5	30-50 PASS	×1.84
E	1.8	3-5.5 low	×1.59
ϵ_f	7.8 %	(range 3-7 %)	×1.02

✓ strength inside journal range (k≈1)

Stress vs crosshead



DIC strain vs crosshead — the compliance gap

